NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Two Surveys of Park Fees

AP Statistics · Unit 1 · Topic 1.12 · B: 2026-10-01 · E: 2026-10-09

[STAR] TODAY'S PROBLEM

Suppose a town studies support for a park fee.

Mail survey: A neutral question is mailed to 500 randomly chosen households. Only 120 return it; 78% of respondents oppose the fee.

Phone survey: The town calls 300 random phone numbers. Of 240 people who answer, 62% oppose after hearing, *Do you support a NEW FEE that would tax park users?*

One council member says 62% must be closer to the truth because more people answered. Another says 78% must be closer because the households were randomly chosen.

Park-fee surveys

	Selected	Replied	Oppose
Mail	500	120	78%
Phone	300	240	62%

FIRST TAKE — before the video (5 min)

Which estimate, if either, would you report to the town? Name one detail behind your choice.

[BOOK] BIAS FAVORS SOME RESPONSES (keep on the page)

Bias systematically favors some responses over others. **Nonresponse bias** is possible when selected people who do not respond differ from those who do; the response rate is responses/selected. **Response bias** can come from confusing or leading wording. A larger sample or response rate does not repair a different design flaw. Match the fix to the source of bias.

Finish after the video:

DOK 1 **(a)** Find each survey's response rate.

DOK 2 **(b)** In two or three sentences, explain one possible bias in each survey. Say which direction each could push opposition, or why its direction is unknown.

DOK 3 **(c)** Has either council member shown which estimate is closer to the town's true opinion? Use one detail from each survey to justify your answer. Choose one survey to improve, and explain how your change addresses its problem.

Frame: I cannot tell which is closer because ____________________. In the mail survey ____________________; in the phone survey ____________________.

Frame: I would change ____________________ because that addresses ____________________.

EXIT — turn this in

One thing the video changed about my first take: ____